

Periodic viscous Burgers as a positive projective heat flow

Global phase portrait, first-mode asymptotics, and a Route-A boundary

HCS-C195 research artifact

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Abstract

For every viscosity $\nu > 0$, circumference $L > 0$, mean $m \in \mathbb{R}$, and $s > 3/2$, we identify the periodic viscous-Burgers semiflow on its fixed-mean H^s leaf with the strictly positive projectivization of a drift-heat semigroup. This gives a global physical-time solution and instant smoothing.

1 Frozen phase leaf and the correct linear state

On $\mathbb{T}_L = \mathbb{R}/L\mathbb{Z}$, consider

$$u_t + uu_x = \nu u_{xx}, \quad X_m^s = \{u \in H^s(\mathbb{T}_L; \mathbb{R}) : L^{-1} \int_{\mathbb{T}_L} u \, dx = m\}. \quad (1)$$

The clock is the source physical time. Define

$$\mathbb{P}_+^{s+1} = \{w \in H^{s+1}(\mathbb{T}_L; \mathbb{R}) : \min w > 0\} / \mathbb{R}_{>0}, \quad \Phi_m([w]) = m - 2\nu \partial_x \log w. \quad (2)$$

The quotient is essential: positive scalar multiples give the same Burgers state. If $V_x = u - m$ and V is the unique zero-mean periodic primitive, then

$$\Phi_m^{-1}(u) = [e^{-V/(2\nu)}]. \quad (3)$$

Sobolev composition makes (2)–(3) smooth inverse maps for $s > 3/2$.

In fixed x , the autonomous linear semigroup is

$$K_t = e^{t(\nu \partial_x^2 - m \partial_x)}, \quad K_t w(x) = (e^{\nu t \partial_x^2} w)(x - mt). \quad (4)$$

Thus pure heat appears after the Galilean coordinate $y = x - mt$; it is not mislabelled as the autonomous fixed-coordinate flow.

Theorem 1 (global projective Cole–Hopf conjugacy). *For all parameters above,*

$$S_t \Phi_m([w]) = \Phi_m([K_t w]), \quad t \geq 0. \quad (5)$$

The cone is invariant, every Burgers orbit is global, and $S_t u$ is smooth for every $t > 0$.

2 Proof mechanism

If $w_t = \nu w_{xx} - mw_x$ and $u = m - 2\nu w_x/w$, direct differentiation gives $u_t + uu_x - \nu u_{xx} = 0$; conversely (3) recovers the same projective linear orbit. The periodic heat kernel preserves strict positivity, proving (5) and global smoothing.

References

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- [2] J. D. Cole, *On a quasi-linear parabolic equation occurring in aerodynamics*, Quart. Appl. Math. 9 (1951), 225–236. JSTOR 43633894.